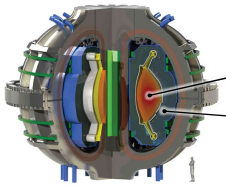
BABY Build A Better Yield blanket

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arpa·e OPEN 2021

Tritium breeding

2030



The ARC concept relies on the Liquid Immersion Blanket for breeding tritium

$$D + T \rightarrow He + n$$

$$Li + n \rightarrow T + He$$

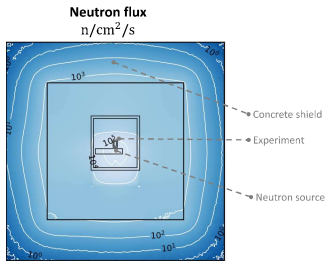
$$\text{Tritium Breeding Ratio} = \frac{\text{Tritium produced}}{\text{Tritium consummed}}$$

Can we test this concept now?

Goals of BABY and LIBRA

- ✅ To demonstrate the concept of Liquid Immersion Blanket
- To improve the Technology Readiness Level of LIB
- To validate numerical models for tritium transport
- To gain knowledge on molten salt behaviour in fusion environments

Neutronics



- Neutronics simulations track the path of neutron particles
- Used to predict tritium generation, neutron flux, heating, production of activated materials...

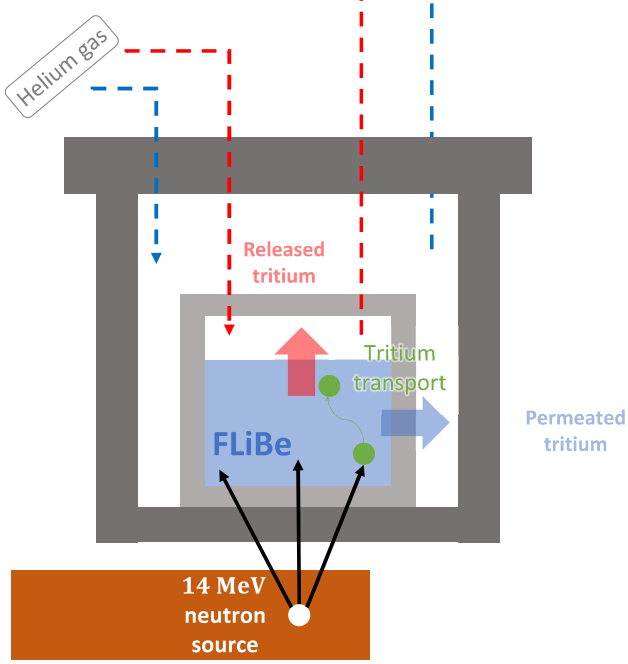


Activation foils

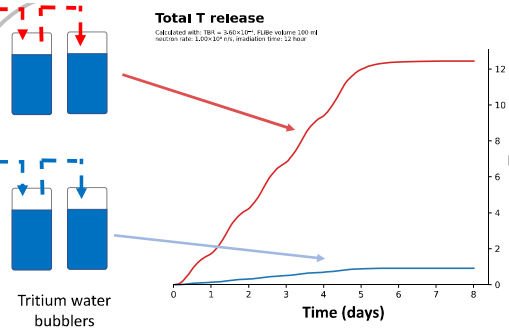
- LIBRA will provide neutronics data for validating codes
- Activation foils and diamond detectors as used to characterize the neutron source and the experiment



Diamond detection



Tritium detection



- Bubblers will absorb 99% of the tritium
- Water samples will be analysed with Liquid Scintillation Counting
- Measurements on BABY will serve as validation data

0D tritium transport model:

$$V\frac{\partial c}{\partial t} = S - A_{\mathrm{p}}\ k_{\mathrm{p}}\ c - A_{\mathrm{r}}\ k_{\mathrm{r}}\ c$$

V: salt volume
c: salt tritium concentration
S: tritium source
A_i: surface area
k_i: mass transport coefficient

BABY specs

Salt volume	$0.1 - 1$ L
TBR	$\approx 10^{-3}$
Temperature	650 °C
Neutron energy	14 MeV

CAD drawings

